

Unit II :: Work and Energy

 **PHY01MX01R04: Mechanics and Properties of Matter** 

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Syllabus

Unit II: Work and Energy

Concepts of work, power and energy; conservative and non-conservative forces; the work-energy theorem; conservative force as the negative gradient of potential energy; the law of conservation of mechanical energy; work done by non-conservative forces; linear restoring force; potential energy curve and equilibria – stable, unstable and neutral.

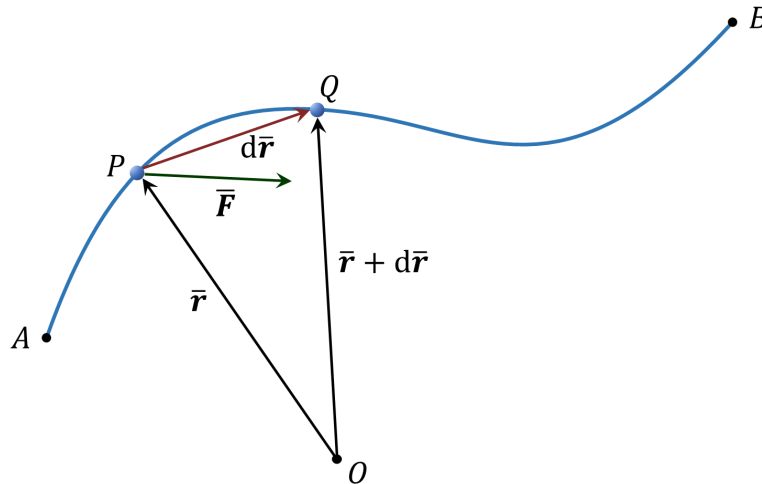
1 Concepts of work, energy and power

Before examining the work-energy theorem and conservation laws in mechanics, we must establish rigorous vector definitions for the foundational quantities of work, energy, and power:

- *Work*: an activity where a force acting on a system causes a displacement.
- *Energy*: the capacity of a system for doing work. A system may possess energy in the form of motion or position.
- *Kinetic energy*: the energy possessed by a system by virtue of its motion.
- *Potential energy*: the energy possessed by a system by virtue of its position or configuration.
- *Power*: the rate at which work is done or energy is used by a system.

1.1 Work

Let us consider a particle of mass m which is moving on an arbitrary path between points A and B . Suppose when the particle is at point P with a position vector $\mathbf{r}$, it is under the influence of some force $\mathbf{F}$ which causes it to get displaced to another point Q with a position vector $\mathbf{r} + d\mathbf{r}$.



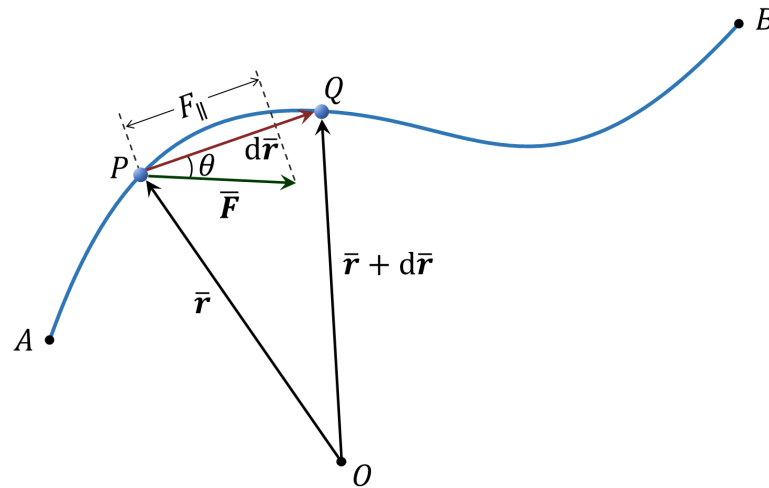
The infinitesimal amount of work done by the force $\mathbf{F}$ in moving the particle from P to Q is

$$dW = \mathbf{F} \cdot d\mathbf{r}$$

Hence, the total work done by the force $\mathbf{F}$ in moving the particle from A to B is

$$W = \int_A^B dW = \int_A^B \mathbf{F} \cdot d\mathbf{r}$$

Using projection of force vector



If the angle θ between $\mathbf{F}$ and $d\mathbf{r}$ is known, then we have

$$dW = F \cos \theta \, dr = F_{\parallel} \, dr$$

where $F_{\parallel} = F \cos \theta$ is the projection of $\mathbf{F}$ along $d\mathbf{r}$. Then we can obtain the total work as

$$W = \int_A^B F \cos \theta \, dr = \int_A^B F_{\parallel} \, dr$$

Using components of force vector

We can write the force vector in component form as follows:

$$\mathbf{F} = \hat{\mathbf{x}}F_x + \hat{\mathbf{y}}F_y + \hat{\mathbf{z}}F_z$$

Also, the infinitesimal displacement is

$$d\mathbf{r} = \hat{\mathbf{x}}dx + \hat{\mathbf{y}}dy + \hat{\mathbf{z}}dz$$

Then the infinitesimal displacement becomes

$$\begin{aligned} dW &= \mathbf{F} \cdot d\mathbf{r} \\ &= (\hat{\mathbf{x}}F_x + \hat{\mathbf{y}}F_y + \hat{\mathbf{z}}F_z) \cdot (\hat{\mathbf{x}}dx + \hat{\mathbf{y}}dy + \hat{\mathbf{z}}dz) \\ \Rightarrow dW &= F_x dx + F_y dy + F_z dz \end{aligned}$$

Therefore, the total work is obtained as

$$\begin{aligned} W &= \int_A^B (F_x dx + F_y dy + F_z dz) \\ \Rightarrow W &= \int_A^B F_x dx + \int_A^B F_y dy + \int_A^B F_z dz \end{aligned}$$

Examples

1. Compute the work done in moving a particle from position $(0, 1, 2)$ to $(5, 6, 8)$ by the force $\mathbf{F} = \hat{\mathbf{x}}(2xy + z^2) + \hat{\mathbf{y}}x^2 + \hat{\mathbf{z}}2xz$.

Solution:

The given force is $\mathbf{F} = \hat{\mathbf{x}}(2xy + z^2) + \hat{\mathbf{y}}x^2 + \hat{\mathbf{z}}2xz$, from which we can identify that

$$F_x = 2xy + z^2, F_y = x^2, F_z = 2xz$$

Also, identifying the coordinates $(0, 1, 2)$ and $(5, 6, 8)$ to be those of the initial and final points A and B respectively, we have the total work done to be

$$\begin{aligned} W &= \int_A^B F_x dx + \int_A^B F_y dy + \int_A^B F_z dz \\ &= \int_A^B (2xy + z^2) dx + \int_A^B x^2 dy + \int_A^B 2xz dz \\ &= \int_A^B (2xy dx + x^2 dy) + \int_A^B (z^2 dx + 2xz dz) \\ &= \int_A^B d(x^2 y) + \int_A^B d(xz^2) \\ &= x^2 y \Big|_A^B + xz^2 \Big|_A^B \\ &= [x^2 y + xz^2]_A^B \\ &= x[x y + z^2]_{x=0, y=1, z=2}^{x=5, y=6, z=8} \\ &= 5 \cdot (5 \cdot 6 + 8^2) - 0(0 \cdot 1 + 2^2) \\ &= 5 \cdot (30 + 64) - 0 \\ &= 5 \times 94 \\ \therefore W &= 470 \text{ units} \end{aligned}$$

1.2 Energy**1.3 Power**

2 Work and energy

2.1 Work-energy theorem

2.2 Potential energy

3 Conservative and non-conservative forces

3.1 Conservative force as a negative gradient of potential energy

3.2 Law of conservation of mechanical energy

3.3 Work done by non-conservative forces

4 Linear restoring force

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5 Potential energy and equilibria

5.1 Potential energy curve

5.2 Types of equilibria

6 General law of conservation of energy

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